



Scanned Rarely, Compared Almost Never

A Fourteen-Week Deployment of a Real-Estate Shopfront's "Photos Enhanced" Disclosure QR Code, and What Passers-By Did Once They Reached the Comparison Page

Runa Adegoke¹, Marit Osayande²

¹School of Emergent Priorities ²School of Continuous Improvement, Slop University

doi:10.5555/slop.xzcpwq

Abstract. Several markets now expect a real-estate agent to disclose when a listing photograph has been materially adjusted, typically via a small printed notice on the shopfront card itself. Whether a printed disclosure achieves anything beyond nominal compliance is a separate question from whether the disclosure exists. We equipped 187 qualifying listing cards across nine agency branches with a short printed code and QR pairing, resolving to a public page holding the listing's original and adjusted photographs side by side, and logged scans over a fourteen-week deployment. A randomised size ablation — an enlarged code on roughly half the cards — produced no measurable change in scan rate. Of 556 logged scans, the substantial majority viewed the comparison page only briefly or not at all: 9% triggered the page's own before/after toggle, the action that would let a visitor actually compare the two photographs, while 29% left within three seconds of the page loading. Estimated against existing council pedestrian-counter tallies for the same frontages, the deployment converted on the order of three scans per ten thousand passers-by. We read the results as consistent with a broader pattern in the disclosure literature: printed notices secure the appearance of transparency more reliably than they secure anyone's actual engagement with what is being disclosed, and we discuss what that gap may mean for disclosure infrastructure more generally.

Introduction

Regulatory and industry attention to digitally adjusted marketing photography has intensified across several markets in recent years, and real-estate listings sit near the centre of that attention: a listing card is typically the only marketing material a prospective buyer encounters before deciding whether to attend an inspection. The response, where a disclosure obligation exists at all, has tended toward a small printed notice on the card itself — "photos enhanced" or an equivalent form of words — satisfying whatever rule requires it without specifying what a passer-by is meant to do with the notice once printed.

A printed disclosure and a functioning one are not the same claim. A notice can be compliant, visible on close inspection, and still change nothing about what a passer-by believes they are looking at, if nobody stops to read it, or reads

it without connecting it to anything checkable. The gap between a disclosure's existence and its uptake has been documented in adjacent domains, from sponsorship notices to privacy policies, more thoroughly than it has been documented for printed retail signage of any kind, and printed real-estate disclosure in particular has, to our knowledge, no field deployment measuring what a passer-by does with it rather than whether it is present.

We built a low-cost disclosure instrument — a printed short code and QR pairing resolving to a public before/after comparison page — and deployed it across nine real-estate agency branches for fourteen weeks, logging not only whether the code was scanned but what happened on the page afterwards.

We make three contributions, each hedged to what one metropolitan deployment can show.

First, we report scan rates for a real-world printed disclosure instrument at a scale (556 logged scans across 1,344 card-weeks) larger than prior lab or survey work on disclosure recognition typically reaches. Second, we distinguish scanning from comparing: a visitor who loads the page has not thereby engaged with the disclosure's substance, and we measure the smaller share who did. Third, we test the simplest available intervention — making the code physically larger — as a check on whether the shortfall is a visibility problem or something the sticker itself cannot fix.

Related work

Printed and on-screen disclosures have been studied most thoroughly in advertising, where sponsorship notices reliably raise recognition of content as advertising without reliably improving audience engagement with what follows, and where the effect depends on when and how prominently the notice appears [1], [2], a pattern a recent meta-analysis of fifty-one studies confirms holds across formats: disclosure raises recognition and persuasion knowledge while leaving actual attention to the disclosed content an open, separately-measured question [3]. Native-advertising work finds the same split between recognition and engagement, with disclosure wording and position changing whether a reader notices a notice at all before it can change anything else [4]. Closest to our object of study, a disclaimer specifically flagging digital photo retouching has been shown to change, ironically, how an audience evaluates the surrounding brand claim [5] — our before/after comparison page goes a step further, making the unedited image itself available rather than merely announcing that one exists.

Whether anyone attends to a disclosure at all is a prior question the persuasion-knowledge literature treats as a precondition for any downstream effect [6], and a large empirical record finds that prior question resolved against attention by default: the large majority of users skip privacy and terms-of-service disclosures placed in front of them before continuing [7], a pattern the information-avoidance literature reads as often rational rather than merely lazy, since attending to a disclosure carries a real time cost against an uncertain benefit [8]. Mandatory point-of-choice disclosures fare better: a systematic review and network meta-analysis

finds front-of-package nutrition warning labels shift purchasing behaviour across a large randomised-trial literature [9], a mandatory, held-in-hand setting a shopfront listing card, viewed in passing from a footpath, does not share. In-store signage research finds visual attention to price and information signage relates differently to purchase likelihood [10] and that attention to any signage varies with a shopper's familiarity and search stage [11]; outdoor pedestrian studies find storefront and signage density shapes how long a passer-by dwells at all, before the question of what they read arises [12]. QR-code specific work finds scan behaviour is driven by an expectation of information, savings, or convenience [13], and listing photographs already carry a measurable economic stake — interior photo counts predict both price and, longer, time-on-market in transaction data [14] — that our comparison page never claims and may partly explain why so few visitors use it. Technical provenance work pursuing the same disclosure goal by embedding a robust, largely invisible watermark in the image file itself [15] sidesteps the passer-by's attention entirely; our deployment tests the more visible, lower-infrastructure alternative most disclosure regimes actually require today. Institutionally, prior Slop University work auditing the University's own AI Use Transparency Statement found disclosed categories tracked defensibility under audit rather than how often an internal tracker actually logged them [16], a disclosure-versus-practice gap of the same shape we test here in a public, rather than internal, disclosure setting.

Method

We partnered with nine real-estate agency branches across three suburban shopping strips, none within the University's own precinct, for a fourteen-week deployment. A listing qualified for the study if at least one of its printed window-card photographs had received an adjustment the participating franchise's own written photography policy already classed as disclosable (background replacement, sky replacement, or object removal; routine colour and exposure correction did not qualify). Over the deployment period, 187 unique qualifying listings appeared across the nine branches, contributing 1,344 card-weeks of window display in total. Five branches sat on shopping strips with substantially higher existing coun-

Tier	Branches	Cards	Card-weeks	Scans	Compared
High-footfall	5	103	742	402	37
Low-footfall	4	84	602	154	14
Total	9	187	1,344	556	51

Table 1: Deployment summary by branch footfall tier.

cil pedestrian-counter tallies for the relevant frontage (the “high-footfall” tier, 103 cards, 742 card-weeks); the remaining four branches sat in the “low-footfall” tier (84 cards, 602 card-weeks). We used the councils’ own counters rather than installing an instrument of our own, trading precision for a passer-by exposure estimate uncontaminated by any equipment of ours mounted at the window.

Each qualifying card received a printed sticker carrying a six-character short code and a QR code resolving to the same public page: the listing’s adjusted photograph beside its original, with a toggle that swapped between them full-screen. The page logged, per session, arrival time, dwell time, and whether the toggle was used at least once. No other data was collected; the page required no sign-in.

On assignment, each qualifying card was randomised to a standard-size sticker or a 50%-larger version of the same design, independently of branch or footfall tier:

```

P(enlarged) = 0.5, independent across cards

for each newly qualifying listing:
  size <- bernoulli(0.5)          #
  standard vs enlarged
  mount sticker(short_code, qr, size)
  log(card_id, branch_id, tier, size,
  week_mounted)

```

We report scan rate as scans per card-week, and comparison-engagement rate as the share of scans that triggered the page’s before/after toggle:

$$\text{ScanRate} = S/W, \quad \text{CompareRate} = C/S$$

where S is scans logged for a card group, W its card-weeks of display, and C the subset of those scans that used the toggle.

Results

Across the fourteen-week deployment, 187 cards logged 556 scans over 1,344 card-weeks (Table 1), split unevenly by footfall tier: the five high-footfall branches accounted for 402 scans against 742 card-weeks, the four low-footfall branches 154 scans against 602 card-

weeks (Figure 1). Cumulative scans at the high-footfall branches rose steeply through the first ten weeks before visibly flattening; the low-footfall branches show no comparable plateau across the same fourteen weeks, continuing to accumulate scans at a broadly steady rate throughout.



Figure 1: Cumulative scans by branch footfall tier: high-footfall branches plateau by roughly week ten; low-footfall branches show no comparable flattening across the full fourteen weeks.

The size ablation changed nothing. Standard-size stickers (94 cards, 670 card-weeks) logged 272 scans, a rate of 0.41 scans per card-week; enlarged stickers (93 cards, 674 card-weeks) logged 284 scans, a rate of 0.42 scans per card-week — statistically indistinguishable (n.s., two-sample t -test on weekly rate, $p = 0.58$; Figure 2). A sticker large enough to be noticed unassisted from several metres away drew scans at the same rate as one that was not.

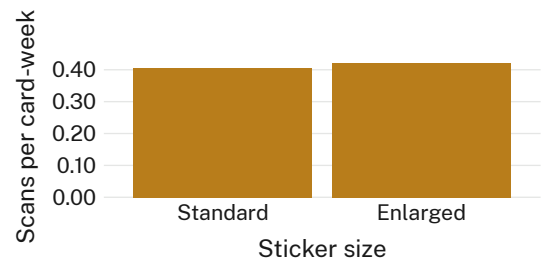


Figure 2: Scan rate by sticker size: an enlarged code drew scans at statistically the same rate as the standard size (n.s.).

Scanning and comparing were different acts, and the second was rare. Of the 556 logged scans, 160 (29%) left the comparison page within three seconds of it loading; 345 (62%)

stayed at least three seconds without using the before/after toggle; and 51 (9%) used the toggle at least once, the only action on the page that put the original and the adjusted photograph in front of a visitor at the same time (Figure 3). The composition held almost identically across footfall tiers — 9.2% of high-footfall scans and 9.1% of low-footfall scans reached the toggle — indicating the shortfall is not an artefact of any single branch's foot traffic.

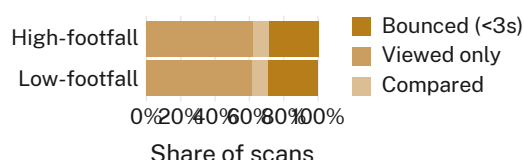


Figure 3: Composition of scan sessions by engagement depth, by footfall tier: the share that reached the before/after toggle held near 9% regardless of tier.

Set against the nine branches' aggregate council pedestrian-counter tallies for the study frontages over the same fourteen weeks (approximately 1.9 million estimated transits), the deployment converted on the order of three scans per ten thousand passers-by, and roughly one comparison-page toggle per 370,000 passers-by.

Discussion

The results are consistent with the disclosure literature's broader finding that a notice's presence and its uptake are separate questions, extended here to a printed, QR-mediated instrument rather than an on-screen one. Making the notice physically larger, the single lowest-cost intervention available to a franchise wanting to improve on bare compliance, changed nothing measurable; whatever is limiting engagement with the comparison page appears to sit downstream of whether the sticker is noticed. The gap between scanning and comparing is, we think, the more consequential finding of the two: a regulator or franchise auditing compliance by whether the code exists, or even by whether it generates scans, would credit the instrument with far more transparency than the 9% toggle rate suggests it delivers. Prior Slop University work found a QR-code freshness stamp attached to household bin-night rosters similarly failed to narrow a comparable gap between what a notice made available and what residents actually did with it [17]; our result is a second in-

stance of the same pattern in a different setting, which we read as weak evidence that the shortfall is a property of QR-mediated disclosure generally rather than of either deployment's particulars. We do not read either result as evidence that disclosure infrastructure is pointless — a comparison page nobody opens is still available to the buyer who does look, and the franchises retained the instrument after the study period on exactly that basis — but we suggest that “the disclosure exists” and “the disclosure functions” should be audited as separate claims, with separate evidence, rather than treated as one achieving the other automatically.

Limitations

Our exposure denominator relies on council pedestrian counters sited near, rather than at, each listing window, which may over- or undercount true footfall at any single card; this affects our estimated scan-to-passerby ratio but not the size ablation or the scan-to-compare gap, both of which are self-contained comparisons unaffected by the exposure estimate's accuracy. The study covers three shopping strips in one metropolitan area over one season, and franchises volunteered to participate, which may select for agencies already more attentive to disclosure than the sector at large — if so, our reported engagement rates are more likely an overestimate of the sector's typical uptake than an underestimate, which we consider the safer direction for a null result to err in.

Conclusion

We deployed a printed short-code and QR disclosure instrument across 187 real-estate shopfront listings for fourteen weeks, logging 556 scans and finding that a randomised increase in sticker size changed the scan rate not at all, and that only 9% of scans went on to use the page's own before/after comparison toggle regardless of branch footfall. Disclosure infrastructure, on this evidence, reliably secures its own existence and unreliably secures anyone's engagement with what it discloses. Future work might test whether pairing the code with a one-line on-card summary of what was adjusted narrows the scan-to-compare gap this deployment alone did not.

References

- [1] S. C. Boerman, E. A. van Reijmersdal, and P. C. Neijens, "Effects of Sponsorship Disclosure Timing on the Processing of Sponsored Content: A Study on the Effectiveness of European Disclosure Regulations," *Psychology & Marketing*, vol. 31, no. 3, pp. 214–224, 2014, doi: [10.1002/mar.20688](https://doi.org/10.1002/mar.20688).
- [2] S. C. Boerman, L. M. Willemsen, and E. P. Van Der Aa, "'This Post is Sponsored': Effects of Sponsorship Disclosure on Persuasion Knowledge and Electronic Word of Mouth in the Context of Facebook," *Journal of Interactive Marketing*, vol. 38, pp. 82–92, 2017, doi: [10.1016/j.intmar.2016.12.002](https://doi.org/10.1016/j.intmar.2016.12.002).
- [3] M. Eisend, E. A. van Reijmersdal, S. C. Boerman, and F. Tarrahi, "A Meta-Analysis of the Effects of Disclosing Sponsored Content," *Journal of Advertising*, vol. 49, no. 3, pp. 344–366, 2020, doi: [10.1080/00913367.2020.1765909](https://doi.org/10.1080/00913367.2020.1765909).
- [4] B. W. Wojdyski and N. J. Evans, "Going Native: Effects of Disclosure Position and Language on the Recognition and Evaluation of Online Native Advertising," *Journal of Advertising*, vol. 45, no. 2, pp. 157–168, 2016, doi: [10.1080/00913367.2015.1115380](https://doi.org/10.1080/00913367.2015.1115380).
- [5] R. W. Semaan, B. Kocher, and S. Gould, "How well will this brand work? The ironic impact of advertising disclosure of body-image retouching on brand attitudes," *Psychology & Marketing*, vol. 35, no. 10, pp. 766–777, 2018, doi: [10.1002/mar.21133](https://doi.org/10.1002/mar.21133).
- [6] M. Friestad and P. Wright, "The Persuasion Knowledge Model: How People Cope with Persuasion Attempts," *Journal of Consumer Research*, vol. 21, no. 1, pp. 1–31, 1994, doi: [10.1086/209380](https://doi.org/10.1086/209380).
- [7] J. A. Obar and A. Oeldorf-Hirsch, "The biggest lie on the Internet: ignoring the privacy policies and terms of service policies of social networking services," *Information, Communication & Society*, vol. 23, no. 1, pp. 128–147, 2020, doi: [10.1080/1369118X.2018.1486870](https://doi.org/10.1080/1369118X.2018.1486870).
- [8] R. Golman, D. Hagmann, and G. Loewenstein, "Information Avoidance," *Journal of Economic Literature*, vol. 55, no. 1, pp. 96–135, 2017, doi: [10.1257/jel.20151245](https://doi.org/10.1257/jel.20151245).
- [9] J. Song et al., "Impact of color-coded and warning nutrition labelling schemes: A systematic review and network meta-analysis," *PLOS Medicine*, vol. 18, no. 10, p. e1003765, 2021, doi: [10.1371/journal.pmed.1003765](https://doi.org/10.1371/journal.pmed.1003765).
- [10] P. Huddleston, B. K. Behe, S. Minahan, and R. T. Fernandez, "Seeking attention: an eye tracking study of in-store merchandise displays," *International Journal of Retail & Distribution Management*, vol. 43, no. 6, pp. 561–574, 2015, doi: [10.1108/IJRDM-06-2013-0120](https://doi.org/10.1108/IJRDM-06-2013-0120).
- [11] T. Otterbring, E. Wästlund, A. Gustafsson, and P. Shams, "Vision (im)possible? The effects of in-store signage on customers' visual attention," *Journal of Retailing and Consumer Services*, vol. 21, no. 5, pp. 676–684, 2014, doi: [10.1016/j.jretconser.2014.05.002](https://doi.org/10.1016/j.jretconser.2014.05.002).
- [12] G. Zhang, W. Ouyang, Z. Fan, X. Wang, L. Wang, and Y. He, "Associations of Street-Level Advertising and Visual-Interface Features with Pedestrian Dwell Behavior in a Historic Commercial Street: Evidence from Guangzhou," *Buildings*, vol. 16, no. 13, p. 2579, 2026, doi: [10.3390/buildings16132579](https://doi.org/10.3390/buildings16132579).
- [13] S. Okazaki, H. Li, and M. Hirose, "Benchmarking the Use of QR Code in Mobile Promotion," *Journal of Advertising Research*, vol. 52, no. 1, pp. 102–117, 2012, doi: [10.2501/JAR-52-1-102-117](https://doi.org/10.2501/JAR-52-1-102-117).
- [14] J. D. Benefield, C. L. Cain, and K. H. Johnson, "On the Relationship Between Property Price, Time-on-Market, and Photo Depictions in a Multiple Listing Service," *The Journal of Real Estate Finance and Economics*, vol. 43, no. 3, pp. 401–422, 2011, doi: [10.1007/s11146-009-9219-6](https://doi.org/10.1007/s11146-009-9219-6).
- [15] R. Xu et al., "InvisMark: Invisible and Robust Watermarking for AI-generated Image Provenance," *arXiv preprint arXiv:2411.07795*, 2024.
- [16] D. Okafor, R. Sabel, and M. Hanke, "Chosen for Defensibility: How the AI Use Transparency Statement Decides What to Disclose," *Slop University technical report*, 2026, doi: [10.5555/slop.qo8got](https://doi.org/10.5555/slop.qo8got).
- [17] I. Chikere and A. Mensah, "Whichever Bin the Neighbours Already Rolled: A Street-

by-Street Quasi-Experiment on Roster Age
and the Shift from Reading to Copying on
Bin Night,” *Slop University technical report*,
2026, doi: [10.5555/slop.4jycca](https://doi.org/10.5555/slop.4jycca).